

Foundation of Programming

Assignment No. 12

PROBLEM STATEMENT:

Write a program in C to accept two numbers from the user and compute:

- a) The smallest common divisor of the two numbers (other than 1, if any)
- b) The Greatest Common Divisor (GCD) of the two numbers

OBJECTIVE:

1. To understand the use of mathematical operations in C.
2. To implement loops and conditional statements.
3. To perform number-based computations.
4. To enhance logical thinking using menu-driven programs.

THEORY:

Divisors and the Greatest Common Divisor (GCD) are fundamental concepts in number theory. The smallest common divisor of two numbers is the smallest positive integer greater than 1 that divides both numbers. The GCD of two numbers is the largest positive integer that divides both numbers without leaving a remainder. In C programming, these operations can be efficiently implemented using loops and the Euclidean algorithm, which repeatedly applies the modulo operation until the remainder becomes zero.

ALGORITHM:

- STEP 1: START
- STEP 2: Declare variables a, b, n, p, i = 1, lcm = 0, gcd
- STEP 3: If $a \geq b$ then $n = a$, else $n = b$
- STEP 4: Multiply a with i and store the result in p.
If $p \% b$ is equal to 0, $\text{lcm} = p$
Else $i = i + 1$
- STEP 5: If lcm is not equal to 0, display lcm
Else repeat step 4
- STEP 6: Run a loop from $i = n$ till i is greater than or equal to 1
If $a \% i$ and $b \% i$ are equal to zero, $\text{gcd} = i$

Else $i = i + 1$
STEP 7: Display gcd
STEP 8: STOP

PLATFORM : Linux

CODE:

```

1  #include <stdio.h>
2
3  // Function to find GCD using Euclidean Algorithm
4  int findGCD(int a, int b) {
5      while(b != 0) {
6          int temp = b;
7          b = a % b;
8          a = temp;
9      }
10     return a;
11 }
12
13 // Function to find smallest common divisor (other than 1)
14 int smallestCommonDivisor(int a, int b) {
15     int i;
16     int min = (a < b) ? a : b;
17
18     for(i = 2; i <= min; i++) {
19         if(a % i == 0 && b % i == 0)
20             return i;
21     }
22     return -1; // No common divisor other than 1
23 }
24
25 int main() {
26     int num1, num2, gcd, scd;
27
28     printf("Enter two numbers: ");
29     scanf("%d %d", &num1, &num2);
30
31     gcd = findGCD(num1, num2);
32     scd = smallestCommonDivisor(num1, num2);
33
34     printf("GCD = %d\n", gcd);
35
36     if(scd == -1)
37         printf("No common divisor other than 1\n");
38     else
39         printf("Smallest Common Divisor (other than 1) = %d\n", scd);
40
41     return 0;
42 }

```

OUTPUT:

```
Enter two numbers: 5 3  
GCD = 1  
No common divisor other than 1
```

CONCLUSION:

Thus, the program to find the smallest common divisor and the Greatest Common Divisor of two numbers was successfully implemented using C programming. This program demonstrates the use of loops, conditional statements, and the Euclidean algorithm.